Deep Learning-Based Fake News Classification Using RNN, LSTM, and Pretrained Word Embeddings (GloVe)

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- Date- 10 May 2026
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Githb Link-https://github.com/SuvamPoudel/AI-and-ML

Abstract
This project presents a comprehensive end-to-end deep learning pipeline for fake news classification using three sequential models: a Simple Recurrent Neural Network (RNN), a Long Short-Term Memory (LSTM) network with trainable embeddings, and an LSTM augmented with frozen pretrained GloVe word embeddings. The dataset used is the True vs. Fake News dataset, a binary-labelled corpus of news articles. Text preprocessing involved lowercasing, contraction expansion, removal of URLs and special characters, stopword elimination, and lemmatisation. All models were trained using the Adam optimiser with binary cross-entropy loss, employing early stopping to prevent overfitting. Results demonstrate that the LSTM with GloVe embeddings (Model 3) achieved the highest test accuracy at 98.55%, outperforming the Simple RNN (94.42%) and trainable-embedding LSTM (92.70%). These findings confirm that pretrained semantic representations substantially improve classification performance on natural language tasks. A real-time prediction interface was also implemented using Gradio, enabling interactive classification of news text across all three models.

Contents

Introduction 3
Dataset 4
Source and Structure 4
Dataset Statistics 4
Preprocessing Summary 4
Methodology 4
Text Preprocessing and Tokenisation 4
Model 1: Simple RNN with Trainable Embedding 5
Model 2: LSTM with Trainable Embedding 5
Model 3: LSTM with Pretrained GloVe Embeddings 5
Training Configuration 5
Experiments and Results 5
Model Performance Summary 5
RNN vs. LSTM Performance 6
Impact of Pretrained GloVe Embeddings 6
Computational Efficiency 6
Error Analysis 7
Misclassification Overview 7
Representative Misclassified Examples 7
Possible Reasons for Errors 7
Suggested Improvements 7
Real-Time Prediction Interface 8
Conclusion and Future Work 8
References 8

Introduction

The rapid proliferation of online misinformation poses a significant challenge to information integrity in contemporary digital societies. Fake news, defined as deliberately fabricated or misleading content presented as factual reporting, has been linked to harmful real-world outcomes including political manipulation, public health misinformation, and social polarisation. Automated detection systems capable of distinguishing credible news from fabricated content are therefore of considerable practical importance.

Traditional machine learning approaches to text classification, such as naive Bayes and support vector machines, rely on bag-of-words or TF-IDF representations that discard sequential and semantic relationships between words. Deep learning architectures, particularly Recurrent Neural Networks (RNNs) and their Long Short-Term Memory (LSTM) variants, are better suited to sequential text data because they maintain hidden state across time steps, enabling the model to capture contextual dependencies within sentences and paragraphs.
This project applies and compares three deep learning models to the binary fake news classification problem. Additionally, the project investigates the effect of initialising word embeddings with pretrained GloVe vectors trained on large external corpora, as opposed to learning embeddings from scratch on the training set alone. The work follows a complete machine learning pipeline covering data loading, preprocessing, tokenisation, model training, evaluation, error analysis, and deployment of a Gradio-based real-time prediction interface.

Dataset

Source and Structure

The dataset used in this project is the True vs. Fake News dataset, a binary-labelled collection of news articles sourced from a CSV file. The dataset contains two primary columns of interest: text, comprising the full body of each news article, and label, a categorical variable taking one of two values: true or false. Labels were numerically encoded as 1 (true) and 0 (fake) for compatibility with the binary classification objective.

Dataset Statistics

The dataset comprises a substantial number of news articles spanning both categories. Class distribution visualisations confirmed that the dataset is approximately balanced between true and fake articles, reducing the risk of class-imbalance bias during training. Text length analysis of the training sequences revealed a wide distribution, with a fixed maximum sequence length of 300 tokens selected to capture the majority of content while maintaining computational tractability.

Preprocessing Summary

Prior to modelling, all text underwent the following preprocessing steps: lowercasing for uniformity; contraction expansion (e.g., "don't" to "do not"); removal of URLs, email addresses, @mentions, hashtags, and numeric characters; removal of punctuation and special characters; stopword removal using the NLTK English stopword list; and lemmatisation using the WordNet Lemmatizer. These steps produced a cleaner, more semantically consistent representation of the raw article text.

Methodology

Text Preprocessing and Tokenisation

Cleaned article text was tokenised using the Keras Tokenizer, configured with a vocabulary limit of 10,000 tokens and an out-of-vocabulary token (<OOV>) for unseen words at inference time. The tokenizer was fit exclusively on the training split to prevent data leakage. The dataset was partitioned into 80% training and 20% testing using stratified random splitting (random_state=42) to preserve class proportions.

Token sequences were padded to a uniform length of 300 tokens using post-padding and post-truncation, consistent with the observed distribution of article lengths in the training data. The resulting padded sequences form the input to all three models.

Model 1: Simple RNN with Trainable Embedding

The first model employs a trainable Embedding layer with an output dimension of 64, followed by a single SimpleRNN layer of 64 units. Two Dropout layers (rate = 0.3) are applied after the recurrent layer and the subsequent Dense(32, activation='relu') layer to reduce overfitting. The output layer uses a single sigmoid-activated neuron appropriate for binary classification.

Model 2: LSTM with Trainable Embedding

The second model shares the same architecture as Model 1 with the exception that the recurrent component is replaced by an LSTM layer of 64 units. LSTMs address the vanishing gradient problem inherent in simple RNNs through gating mechanisms: the input gate, forget gate, and output gate collectively regulate the flow of information through the cell state, enabling the capture of longer-range dependencies in news text.

Model 3: LSTM with Pretrained GloVe Embeddings

The third model replaces the trainable Embedding layer with a fixed pretrained embedding matrix constructed from the GloVe (Global Vectors for Word Representation) model trained on Wikipedia and Gigaword corpora, specifically the 50-dimensional variant (glove-wiki-gigaword-50). A zero-initialised matrix of shape (vocab_size, 50) was populated by looking up each word in the vocabulary against the GloVe index; words absent from GloVe retained zero vectors. The embedding layer was set to non-trainable (trainable=False) to preserve the pretrained semantic structure. The mask_zero=True argument was set to ensure padding tokens are ignored during LSTM computation.

Training Configuration

All three models were compiled using the Adam optimiser and binary cross-entropy loss, with accuracy as the primary evaluation metric. Training was conducted for up to 20 epochs with a batch size of 64 and a 10% validation split drawn from the training data. An EarlyStopping callback monitored validation loss with a patience of 3 epochs, restoring the best weights upon termination to guard against overfitting.

Experiments and Results

Model Performance Summary

Table 1 summarises the test accuracy achieved by each model on the held-out 20% test set following training with early stopping.

Model	Architecture	Embedding	Test Accuracy

Model 1	Simple RNN	Trainable (64-dim)	94.42%
Model 2	LSTM	Trainable (64-dim)	92.70%
Model 3	LSTM + GloVe	Frozen GloVe (50-dim)	98.55%

Table 1. Test Accuracy Comparison Across Models

RNN vs. LSTM Performance

Model 1 (Simple RNN) achieved a test accuracy of 94.42%, demonstrating that even a simple recurrent architecture can extract meaningful sequential patterns from news text. However, Model 2 (LSTM) achieved 92.70%, marginally below the Simple RNN. This counterintuitive result may be attributable to the LSTM's greater parameter count requiring more data to converge effectively under the given epoch and early stopping configuration, or to a degree of overfitting that the validation-based early stopping did not fully prevent. Training history plots revealed that the Simple RNN converged more steeply and stabilised earlier, while the LSTM exhibited slightly more oscillation in validation loss.

The fundamental architectural advantage of LSTM over Simple RNN lies in its gating mechanism, which enables the retention of long-term dependencies while selectively forgetting irrelevant information. For longer news articles, this property is theoretically advantageous; however, under the constrained training conditions of this project, the Simple RNN remained competitive.

Impact of Pretrained GloVe Embeddings

Model 3 (LSTM + GloVe) substantially outperformed both trainable-embedding models, achieving 98.55% test accuracy. This 3.85-percentage-point improvement over Model 2 demonstrates the significant benefit of initialising embeddings with rich, pre-learned semantic representations. GloVe embeddings encode co-occurrence statistics from billions of tokens across Wikipedia and Gigaword, imbuing each word vector with contextual associations that a model trained on the comparatively small news dataset alone cannot acquire. This semantic grounding is particularly valuable for fake news detection, where subtle linguistic cues and word associations carry important signals.

Embedding coverage analysis indicated that the majority of in-vocabulary words were found in the GloVe index, providing strong semantic initialisation for the bulk of the token space. Words absent from GloVe retained zero vectors, which may slightly penalise performance on domain-specific or rare terminology.

Computational Efficiency

Training was conducted on Google Colab using GPU acceleration. Model 1 (Simple RNN) converged fastest owing to its simpler architecture and fewer trainable parameters. Model 2 (LSTM) required slightly longer per-epoch computation due to the LSTM's gating operations. Model 3 (LSTM + GloVe) incurred additional one-time overhead for loading and mapping the GloVe embeddings but trained efficiently at runtime given the non-trainable embedding layer, which reduced the parameter optimisation burden during backpropagation.

Error Analysis

Misclassification Overview

Post-evaluation error analysis was conducted on Model 3 (LSTM + GloVe), the best-performing model. Despite its high accuracy, the model misclassified a subset of test articles. These errors were categorised into false positives, where true news was predicted as fake, and false negatives, where fake news was predicted as true.

Representative Misclassified Examples

Three representative misclassifications from Model 3 were examined. These examples revealed the following patterns:

- False Positive (True predicted as Fake): A genuine news article containing sensational or hyperbolic language was incorrectly flagged as fake. The model likely associated the dramatic phrasing with patterns it had learned from fabricated content.
- False Negative (Fake predicted as True): A fabricated article written in a factual, measured tone with credible-sounding references evaded detection. The absence of obvious linguistic markers of deception caused the model to assign a high probability of authenticity.
- Ambiguous Case: A third example contained mixed signals, incorporating factual elements alongside misleading framing. This class of article is inherently difficult to classify and represents a broader limitation of text-level classification without external fact-checking.

Possible Reasons for Errors

Several factors contribute to misclassification in this setting. First, ambiguous linguistic signals arise when articles adopt the stylistic conventions of the opposing class, either through deliberate mimicry or coincidental overlap. Second, truncation at 300 tokens may cause the model to lose important contextual information present in the latter portions of long articles. Third, vocabulary coverage gaps in GloVe mean that domain-specific or neologistic terms receive zero-vector representations, weakening the semantic signal for articles containing such vocabulary. Fourth, the model operates on surface-level text features and lacks access to external knowledge for factual verification.

Suggested Improvements

To improve classification performance, several directions warrant investigation. Fine-tuning the GloVe embeddings (trainable=True) rather than freezing them may allow the model to adapt pretrained representations to the specific linguistic patterns of the news domain. Bidirectional LSTM layers, which process sequences in both forward and backward directions, could capture richer contextual dependencies. Incorporating an attention mechanism would allow the model to focus on the most discriminative regions of an article. At the architectural level, transformer-based models such as BERT or RoBERTa, which have demonstrated state-of-the-art performance on numerous NLP benchmarks, could provide substantially stronger baselines. Hyperparameter optimisation, including learning rate scheduling, increased LSTM units, and regularisation tuning, could further improve convergence.

Real-Time Prediction Interface

A real-time prediction interface was implemented using Gradio, fulfilling the optional extra-credit component of the assignment. The interface accepts free-text news article input, applies the same preprocessing and tokenisation pipeline used during training, and returns classification predictions from all three models simultaneously. Each prediction is displayed alongside the estimated fake news probability, enabling comparative inspection of model behaviour on user-supplied text.

The interface includes four pre-loaded example inputs, ranging from clearly credible news to obviously fabricated content, facilitating immediate demonstration. The Gradio application was deployed with public sharing enabled via the share=True parameter, producing a shareable URL accessible during the active runtime session. This deployment approach demonstrates a lightweight and accessible route to serving deep learning models for natural language processing tasks without requiring dedicated server infrastructure.

Conclusion

This project successfully implemented and evaluated three deep learning architectures for binary fake news classification. The experimental results establish a clear performance hierarchy: the LSTM with pretrained GloVe embeddings (98.55%) outperformed both the Simple RNN (94.42%) and the LSTM with trainable embeddings (92.70%). These results underscore the practical value of transfer learning in NLP, demonstrating that pretrained word representations encode semantic structure that improves downstream classification even when the embedding layer is held frozen.

The project also highlights the trade-off between model complexity and performance. The Simple RNN, despite its relative simplicity, achieved competitive accuracy, suggesting that for well-preprocessed binary classification tasks with balanced classes, even lightweight recurrent models can perform strongly. The LSTM's theoretical advantage over Simple RNN in capturing long-range dependencies did not translate into improved accuracy under the training conditions of this experiment, though this outcome may differ with larger datasets or longer training budgets.

Limitations of the current approach include the reliance on surface-level textual features without access to external knowledge bases for factual verification, the fixed 300-token truncation which discards information from long articles, and the absence of ensemble or attention-based mechanisms. Future work should explore transformer-based architectures such as BERT, incorporate article metadata such as source credibility or publication date, and investigate cross-domain generalisation across different news datasets. The Gradio deployment demonstrates a viable pathway for real-world application of the trained models, and future iterations could incorporate model updating pipelines to maintain classification accuracy as news language evolves over time.

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